

Hide menu

2.1: Introduction to the course

✓ Reading: Frequently Asked Questions10 min

✓ Reading: Course Resources10 min

✓ Reading: How to Use Discussion Forums10 min

Discussion Prompt: Introduce yourself10 min

✓ Reading: Earn a Course Certificate10 min

✓ Reading: Are you interested in earning an online MSEE degree?10 min

2.1.1: Welcome to the course!

2.1.2: Predict/correct mechanism of sequential probabilistic inference

2.1.3: The Kalman-filter gain factor

2.1.4: Summarizing the six steps of generic sequential probabilistic inference

2.1.5: Deriving the three linear Kalman-filter prediction steps

2.1.6: Deriving the three linear Kalman-filter correction steps

2.1.7: Summary of "Deriving the linear Kalman filter" module plus next steps

Are you interested in earning an online MSEE degree?

You may be interested to know that the University of Colorado at Boulder (CU-Boulder) offers a completely online Master of Science in Electrical Engineering (MSEE) degree program (<https://www.colorado.edu/ecee/msee>). Any work that you have already completed in the Coursera non-degree certificate “Applied Kalman Filtering” can be transferred into this accredited graduate-degree-earning program.

The primary additional requirements of the CU-Boulder program beyond the non-degree Coursera course(s) in which you have already enrolled are:

- To receive credit at CU-Boulder, you must pay tuition to CU-Boulder. The tuition fees for the online MSEE are substantially lower than for an in-person MSEE. There are no financial-aid possibilities.
- You must complete all graded assessments for each Coursera course
- You must complete an online two-hour final examination for each Coursera course. (These are similar to a long weekly quiz and are designed to assesses your mastery of the content for the entire course; each of my courses also has a practice exam for you to attempt before taking the final exam.)

There are further requirements on acceptable grades and so forth, and these are listed on the CU-Boulder website. Note that learners enrolled in the CU-Boulder graduate-degree-earning program benefit from access to CU-Boulder teaching assistants who hold regular "office hours" to help answering questions related to course content.

For each of the courses in the Applied Kalman Filtering specialization, you can receive a varying amount of graduate credit. The following list gives the CU-Boulder course number and amount of credit for each course. The MSEE degree requires a total of 30 credits.

- ECEA 5850 Kalman-filter boot camp and state-estimation application (0.8 credits)
- ECEA 5851 Kalman filter deep dive and target-tracking application (0.8 credits)
- ECEA 5852 Nonlinear Kalman filters, parameter-estimation application (0.8 credits)
- ECEA 5853 Particle filters and navigation application (0.8 credits)

Beyond these courses, CU-Boulder offers specializations in embedded systems; power electronics; photonics and optics; and has a roadmap of planned specializations in control and communications; electromagnetics, RF, microwaves, and remote sensing; and computer engineering. The course list is strong and growing.

If this MSEE opportunity interests you, I recommend that you take some time to explore the CU-Boulder website for the program. You can request more information through <https://www.coursera.org/degrees/msee-boulder#request-info>. If you have questions, please reach out to msee-mooc@colorado.edu. Note that I am only the content provider for my courses and will not be able to help you with admissions questions.

Go to next item

✓ Completed

